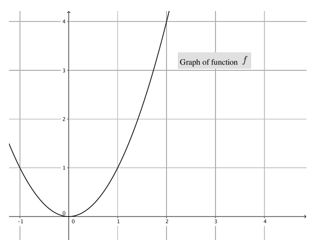
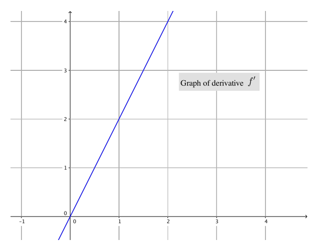
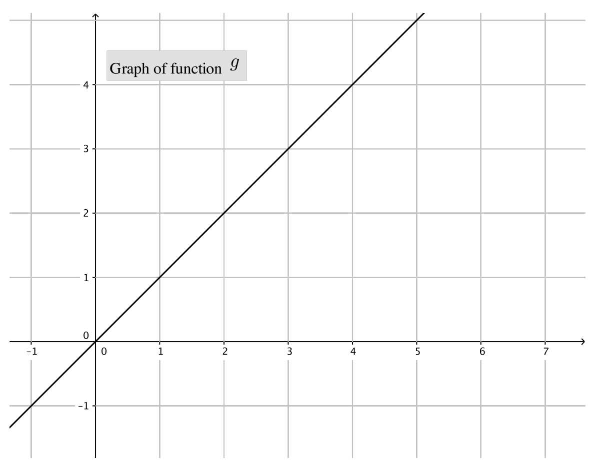
Problem 4

Problem 4

Problem 1 *Use the given graphs of f and g and their accompanying derivatives to answer the following questions.*



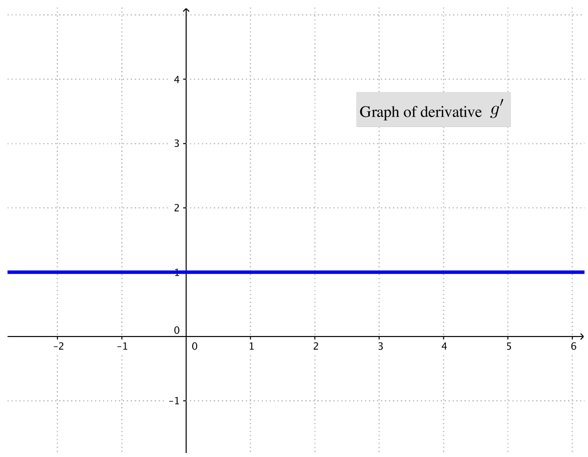




(a) *Write an equation for the tangent line to f at $x = 2$.*

Explanation. $f(2) = 4, \quad f'(2) = 4, \quad y - 4 = 4(x - 2), \quad y = 4x - 4$

(b) *Draw the graph of g'*



Explanation.

(c) *Compute the value of the* $(5f+3g)'(2)$.

Explanation.

$$\begin{aligned}(5f+3g)'(2) &= 5\cdot f'(2)+3\cdot g'(2)\\ &= 5\cdot 4+3\cdot 1\\ &= 20+3=23.\end{aligned}$$

(d) *Find the equation of the tangent line to the graph of* $(5f+3g)$ *at the point where* $x=2$.

Explanation.

$$\begin{aligned}(5f+3g)(2) &= 5\cdot f(2)+3\cdot g(2)\\ &= 5\cdot 4+3\cdot 2\\ &= 20+6=26\end{aligned}$$

$$\begin{aligned}y-26 &= 23(x-2)\\ y &= 23x-20\end{aligned}$$

(e) *Use the given graph of* f' *and* g' *to find the following:*

(i) *A formula for* f'

Explanation. $f'(x)=2x$. The graph shows a linear function through the point $(0,0)$ and $(1,2)$.

(ii) *A formula for* g'

Explanation. $g'(x)=1$. The graph shows a linear function with slope 0 and through the point $(0,1)$.

(f) *Find the expression for* $(5f+3g)'(x)$.

Explanation.

$$\begin{aligned}(5f+3g)'(x) &= 5\cdot f'(x)+3\cdot g'(x)\\ &= 5\cdot 2x+3\cdot 1\\ &= 10x+3\end{aligned}$$